

**DOROZHNY, Russian Federation**

WMO: 248260

Lat: **61.446N**Lon: **114.361E**Elev: **354**StdP: **97.15**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p) |
| 1             | -43.7      | -41.4 | -45.6                         | 0.0 | -43.0 | -43.7 | 0.1 | -41.1 | 5.4                   | -17.3 | 5.1 | -18.7 | 1.5                   | 230  | N/A |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.6                   | 29.1            | 17.7 | 27.1 | 16.9 | 25.0 | 16.0 | 19.0                | 26.4 | 18.1 | 25.1 | 17.1 | 23.3 | 2.3                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.5                            | 12.2 | 21.2 | 15.4 | 11.4 | 20.5 | 14.4 | 10.7 | 19.7 | 55.7          | 26.5 | 52.7 | 25.3 | 49.4 | 23.3 | 22.3              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |    |    |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|----|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |    |    |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         | DB | WB |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |    |    |
| (4) <b>5.4</b>    | <b>4.9</b> | <b>4.3</b> | <b>-45.2</b>               | <b>32.3</b> | <b>3.1</b>         | <b>1.7</b> | <b>-47.4</b>                                       | <b>33.5</b> | <b>-49.2</b> | <b>34.5</b> | <b>-51.0</b> | <b>35.5</b> | <b>-53.2</b> | <b>36.7</b> |    |    |
| (5)               |            |            | <b>-44.7</b>               | <b>20.6</b> | <b>2.5</b>         | <b>1.2</b> | <b>-46.5</b>                                       | <b>21.5</b> | <b>-48.0</b> | <b>22.2</b> | <b>-49.4</b> | <b>22.8</b> | <b>-51.2</b> | <b>23.7</b> |    |    |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -4.9  | -28.0 | -23.8 | -12.1 | -2.6  | 6.0   | 15.3  | 16.7  | 14.4  | 5.5   | -5.2  | -20.2 | -26.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 17.36 | 8.54  | 8.47  | 7.73  | 5.43  | 4.67  | 4.41  | 3.84  | 3.66  | 4.31  | 5.56  | 8.02  | 8.88  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 5993  | 1178  | 946   | 684   | 378   | 141   | 7     | 2     | 8     | 146   | 469   | 907   | 1128  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 8538  | 1437  | 1179  | 942   | 628   | 382   | 109   | 77    | 128   | 386   | 728   | 1157  | 1386  | (9)  |
| (10) |                                                                                  | CDD10.0     | 551   | 0     | 0     | 0     | 0     | 18    | 167   | 210   | 146   | 9     | 0     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 54    | 0     | 0     | 0     | 0     | 1     | 19    | 27    | 7     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 818   | 0     | 0     | 0     | 0     | 9     | 303   | 372   | 132   | 3     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 215   | 0     | 0     | 0     | 0     | 1     | 81    | 109   | 23    | 1     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.1   | 2.0   | 2.1   | 2.3   | 2.5   | 2.5   | 2.0   | 1.9   | 1.8   | 2.0   | 2.1   | 2.0   | 2.0   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 427   | 20    | 16    | 18    | 21    | 35    | 50    | 63    | 59    | 45    | 41    | 34    | 23    | (15) |
| (16) |                                                                                  | PrecMax     | 569   | 42    | 31    | 30    | 50    | 73    | 93    | 130   | 107   | 87    | 70    | 63    | 45    | (16) |
| (17) |                                                                                  | PrecMin     | 338   | 10    | 7     | 4     | 8     | 22    | 24    | 19    | 28    | 14    | 15    | 11    | 8     | (17) |
| (18) |                                                                                  | PrecStd     | 52    | 8     | 6     | 6     | 9     | 12    | 16    | 27    | 20    | 18    | 15    | 12    | 9     | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | -9.0  | -4.3  | 7.4   | 11.9  | 24.3  | 31.6  | 32.3  | 29.2  | 21.8  | 8.5   | 1.9   | -3.0  | (19) |
| (20) |                                                                                  |             | MCWB  | -9.6  | -6.2  | 2.0   | 4.7   | 12.7  | 18.2  | 19.4  | 18.1  | 13.4  | 4.4   | 1.1   | -3.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | -11.8 | -7.2  | 4.1   | 9.7   | 20.8  | 29.0  | 29.6  | 26.7  | 17.5  | 6.1   | -2.2  | -6.9  | (21) |
| (22) |                                                                                  |             | MCWB  | -12.4 | -8.3  | 0.1   | 3.7   | 10.8  | 16.8  | 18.8  | 17.2  | 10.5  | 3.4   | -2.8  | -7.5  | (22) |
| (23) |                                                                                  | 5%          | DB    | -13.6 | -9.7  | 1.1   | 7.8   | 18.1  | 26.8  | 27.3  | 24.7  | 15.2  | 4.1   | -5.1  | -9.6  | (23) |
| (24) |                                                                                  |             | MCWB  | -14.1 | -10.6 | -2.0  | 2.3   | 9.3   | 16.0  | 17.5  | 16.0  | 9.4   | 1.7   | -5.9  | -10.1 | (24) |
| (25) |                                                                                  | 10%         | DB    | -15.8 | -11.9 | -1.1  | 5.9   | 15.1  | 24.6  | 25.0  | 22.5  | 12.6  | 2.3   | -7.7  | -12.5 | (25) |
| (26) |                                                                                  |             | MCWB  | -16.2 | -12.7 | -3.4  | 1.3   | 7.5   | 15.1  | 16.6  | 15.5  | 8.2   | 0.5   | -8.2  | -12.9 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | -9.9  | -6.0  | 2.5   | 5.4   | 13.5  | 19.3  | 20.6  | 19.2  | 13.7  | 5.4   | 1.3   | -3.5  | (27) |
| (28) |                                                                                  |             | MCDWB | -9.3  | -4.6  | 6.8   | 10.2  | 23.1  | 28.6  | 28.9  | 25.8  | 19.4  | 7.9   | 1.7   | -3.1  | (28) |
| (29) |                                                                                  | 2%          | WB    | -12.3 | -8.3  | 0.3   | 4.0   | 11.2  | 18.0  | 19.5  | 18.2  | 11.9  | 3.8   | -2.8  | -7.3  | (29) |
| (30) |                                                                                  |             | MCDWB | -11.9 | -7.3  | 3.7   | 9.0   | 19.4  | 26.9  | 27.2  | 24.5  | 15.8  | 5.6   | -2.2  | -6.9  | (30) |
| (31) |                                                                                  | 5%          | WB    | -14.0 | -10.6 | -1.7  | 2.7   | 9.7   | 17.1  | 18.6  | 17.1  | 10.0  | 2.0   | -5.9  | -10.0 | (31) |
| (32) |                                                                                  |             | MCDWB | -13.6 | -9.7  | 0.8   | 7.0   | 17.0  | 25.1  | 25.6  | 22.7  | 13.6  | 3.6   | -5.2  | -9.5  | (32) |
| (33) |                                                                                  | 10%         | WB    | -16.2 | -12.7 | -3.3  | 1.5   | 8.0   | 16.0  | 17.7  | 16.0  | 8.8   | 0.5   | -8.2  | -12.9 | (33) |
| (34) |                                                                                  |             | MCDWB | -15.8 | -11.9 | -1.1  | 5.5   | 14.8  | 22.5  | 23.8  | 21.2  | 12.5  | 2.2   | -7.7  | -12.5 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 7.0   | 9.0   | 11.0  | 10.6  | 10.5  | 12.2  | 11.6  | 12.0  | 9.1   | 6.4   | 7.4   | 7.1   | (35) |
| (36) |                                                                                  |             | MCDBR | 7.0   | 8.5   | 10.6  | 11.9  | 15.6  | 15.9  | 15.5  | 16.7  | 14.0  | 7.9   | 8.8   | 8.5   | (36) |
| (37) |                                                                                  |             | MCWBR | 6.7   | 7.8   | 7.6   | 7.7   | 8.1   | 6.7   | 6.2   | 7.7   | 7.7   | 5.7   | 8.8   | 8.4   | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 6.8   | 8.6   | 9.6   | 10.3  | 14.2  | 13.8  | 13.6  | 13.4  | 12.0  | 7.2   | 8.5   | 8.7   | (38) |
| (39) |                                                                                  |             | MCWBR | 6.5   | 7.8   | 7.1   | 7.1   | 8.0   | 6.3   | 6.2   | 6.8   | 7.2   | 5.7   | 8.5   | 8.6   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.231 | 0.268 | 0.296 | 0.330 | 0.358 | 0.398 | 0.437 | 0.411 | 0.324 | 0.283 | 0.228 | 0.195 | (40)  |      |
| (41) |                                                                                  | taud        | 1.951 | 2.047 | 2.082 | 2.129 | 2.321 | 2.322 | 2.203 | 2.238 | 2.471 | 2.334 | 1.990 | 1.875 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 454   | 688   | 807   | 854   | 862   | 828   | 777   | 764   | 795   | 691   | 451   | 317   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 42    | 76    | 105   | 123   | 111   | 114   | 126   | 112   | 73    | 57    | 39    | 26    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.39  | 1.33  | 3.03  | 4.88  | 5.21  | 5.75  | 5.14  | 4.00  | 2.47  | 1.13  | 0.50  | 0.20  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.02  | 0.08  | 0.15  | 0.29  | 0.36  | 0.43  | 0.41  | 0.24  | 0.24  | 0.12  | 0.05  | 0.01  | (45)  |      |